The Campus Notice Broadcaster

Week 8 Coding Assignment - Question 5

Design: NoticeBoard manages students and posts notices. A Student has one department and one or more
NotificationChannel preferences. NotificationChannel is realized by EmailChannel, SmsChannel, and AppChannel.
Java solution:

```
import java.util.*;
interface NotificationChannel {String name();void send(Student s,Notice n);}
class EmailChannel implements NotificationChannel {public String name(){return "Email";}public
void send(Student s,Notice n){System.out.println("[Email -> "+s.name+"] "+n.title);}}
class SmsChannel implements NotificationChannel {public String name(){return "SMS";}public
void send(Student s,Notice n){System.out.println("[SMS -> "+s.name+"] "+n.title);}}
class AppChannel implements NotificationChannel {public String name(){return "App";}public
void send(Student s,Notice n){System.out.println("[App -> "+s.name+"] "+n.title);}}
class Student {final String name,department;final List<NotificationChannel>
channels;Student(String n,String d,List<NotificationChannel>
c){name=n;department=d;if(c.isEmpty())throw new IllegalArgumentException("Choose at least one
channel");channels=new ArrayList<>(c);}}
class Notice {final String title;final Set<String> targets;Notice(String t,Set<String>
d){if(t==null||t.trim().isEmpty())throw new IllegalArgumentException("Notice title is
required");if(d==null||d.isEmpty())throw new IllegalArgumentException("At least one target
department is required");title=t;targets=new LinkedHashSet<>(d);}}
class NoticeBoard {final List<Student> students=new ArrayList<>();void add(Student
s){students.add(s);}void post(Notice n){
System.out.println("Notice '"+n.title+"' posted to "+String.join(", ",n.targets)+".");
for(Student s:students)if(n.targets.contains(s.department))for(NotificationChannel
c:s.channels)c.send(s,n);
}}
public class Main {public static void main(String[] args){
NoticeBoard b=new NoticeBoard();b.add(new Student("Asha","CSE",Arrays.asList(new
EmailChannel(),new AppChannel())));b.add(new Student("Ravi","ECE",Arrays.asList(new
SmsChannel())));
b.post(new Notice("Lab Closed Tomorrow",new LinkedHashSet<>(Arrays.asList("CSE"))));
b.post(new Notice("Fee Deadline Extended",new LinkedHashSet<>(Arrays.asList("CSE","ECE"))));
try{b.post(new Notice("Sports Day",Collections.emptySet()));}catch(IllegalArgumentException
e){System.out.println("Cannot post notice: "+e.getMessage());}
} }
```

Expected output includes email and app messages for Asha, SMS messages for Ravi, and a rejection for an empty target list.
NoticeBoard depends on the NotificationChannel abstraction; adding WhatsAppChannel does not change posting logic.